

Derivation of the average energy of a Planck resonator

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Introduction

This document derives the average energy of a quantum harmonic oscillator in the context of Planck's hypothesis for blackbody radiation.

Derivation

We consider a quantum harmonic oscillator of frequency ν , whose allowed energy levels are

$$E_n = n h \nu \quad (n = 0, 1, 2, \dots).$$

1. Partition Function

In the canonical ensemble at temperature T , the probability of occupying the energy level n is given by

$$P_n = \frac{e^{-E_n/(k_B T)}}{Z},$$

where Z is the *partition function*:

$$Z = \sum_{n=0}^{\infty} e^{-\frac{n h \nu}{k_B T}}.$$

Define the shorthand

$$x = e^{-\frac{h\nu}{k_B T}}.$$

Then,

$$Z = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad (\text{since } |x| < 1 \text{ for } T > 0).$$

2. Average Energy

The average energy $\langle E \rangle$ is given by

$$\langle E \rangle = \sum_{n=0}^{\infty} E_n P_n = \frac{1}{Z} \sum_{n=0}^{\infty} (n h \nu) e^{-\frac{n h \nu}{k_B T}}.$$

Factor out $h\nu$:

$$\langle E \rangle = \frac{h\nu}{Z} \sum_{n=0}^{\infty} n x^n.$$

We use the known series result:

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2}.$$

Hence,

$$\langle E \rangle = \frac{h\nu}{Z} \frac{x}{(1-x)^2}.$$

Recall that $Z = \frac{1}{1-x}$, so

$$\frac{1}{Z} = 1 - x.$$

Thus,

$$\langle E \rangle = h\nu (1-x) \frac{x}{(1-x)^2} = h\nu \frac{x}{1-x}.$$

Finally, substitute back $x = e^{-\frac{h\nu}{k_B T}}$:

$$\langle E \rangle = h\nu \frac{e^{-\frac{h\nu}{k_B T}}}{1 - e^{-\frac{h\nu}{k_B T}}}.$$

Often, to get the standard form, multiply numerator and denominator by $e^{\frac{h\nu}{k_B T}}$:

$$\langle E \rangle = \frac{h\nu}{e^{\frac{h\nu}{k_B T}} - 1}.$$